

Real Time Project Report

on

**“WIRELESS SENSOR NETWORK FOR
AGRICULTURE”**

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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(UGC Autonomous, Approved by AICTE and Affiliated to JNTU, Hyderabad)

Accredited by NBA and NAAC with A Grade

Bachupally, Hyderabad – 500090

2024-25

DECLARATION

We hereby declare that the work described in this report, entitled “**WIRELESS SENSOR NETWORK FOR AGRICULTURE**” which is being submitted by us in partial fulfilment for the award of the degree of **Bachelor of Technology** in the department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women (UGC Autonomous)** affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad –500085 is the result of original work carried out by us under the guidance of **Mr. SAIKUMARA TARA**. This work has not been submitted for any Degree/Diploma of this or any other institute/university to the best of our knowledge and belief.

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Certificate

This is to certify that the mini project report, entitled **“WIRELESS SENSOR NETWORK FOR AGRICULTURE”** is a record of Bonafide work by **MamathaKumari(23WH1A0460), Y. Vinayalatha(23WH1A0414), Likhitha Reddy(23WH1A0406), K. Nandini(23WH1A0457)**, in partial fulfilment for the award of the degree of **Bachelor of Technology** in the department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women(UGC Autonomous)**, affiliated to **Jawaharlal Nehru Technological University Hyderabad, Kukatpally, Hyderabad – 500085**.

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ABSTRACT

(WSNs) in agriculture to monitor key environmental parameters such as soil moisture, temperature, and humidity. By deploying a network of sensors across agricultural fields, real-time data is transmitted wirelessly to a central system for analysis. This enables farmers to optimize irrigation, improve crop health, and conserve resources, ultimately This project explores the use of **Wireless Sensor Networks** enhancing productivity and sustainability in farming practices **Wireless Sensor Networks (WSNs)** are systems of interconnected sensors used to monitor environmental conditions like soil moisture, temperature, and humidity. In agriculture, these sensors are deployed across fields to collect real-time data for better crop and resource management. Data from sensor nodes is transmitted wirelessly to a central system for analysis and action. WSNs use technologies like Zigbee, LoRa, or Wi-Fi depending on the range and power needs. They enable precision farming, reduce resource waste, and support sustainable agricultural practices. In conclusion, **Wireless Sensor Networks** enhance agricultural efficiency through real-time monitoring and data-driven decision-making. They help optimize resource use, improve yields, and support sustainable farming practices. Despite some challenges, WSNs are a key technology in the future of smart agriculture.

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1. INTRODUCTION

Wireless Sensor Networks (WSNs) are transforming agriculture by enabling real-time, data-driven monitoring of critical environmental factors such as soil moisture, temperature, humidity, and light intensity. These networks consist of interconnected, low-cost sensors that wirelessly transmit field data, allowing farmers to make precise decisions about irrigation, fertilization, and pest control. By replacing traditional guesswork with accurate, automated insights, WSNs enhance crop yields while reducing water waste, energy consumption, and chemical usage.

They support sustainable farming practices, helping address challenges like climate change and resource scarcity. With remote access via mobile or cloud platforms, farmers can monitor conditions in real time and automate responses, such as triggering irrigation systems. Despite challenges like power efficiency and network scalability, advancements in solar-powered sensors, AI analytics, and IoT integration are paving the way for smarter, more efficient agriculture. As technology evolves, WSNs will play an increasingly vital role in building resilient, eco-friendly farming systems for the future.

Wireless Sensor Networks (WSNs) are emerging as a game-changing technology in agriculture, offering unprecedented visibility into crop and environmental conditions through distributed, self-organizing sensor nodes. These intelligent systems go beyond basic monitoring by enabling predictive analytics for crop diseases, yield forecasting, and automated farm machinery control. Modern WSNs incorporate advanced features like edge computing for local data processing, reducing latency and bandwidth requirements while maintaining data accuracy. They can integrate with satellite imagery and drone-based remote sensing to create comprehensive digital twins of agricultural fields.

The latest innovations include biodegradable sensors for temporary deployment and energy-harvesting technologies that power nodes through soil microbial activity or plant electrophysiology. WSNs also facilitate compliance with precision agriculture standards and sustainable farming certifications by providing verifiable data on resource usage. As cybersecurity becomes crucial in aggrotech, next-generation WSNs are incorporating blockchain-based data integrity checks and encrypted mesh networking protocols.

2. COMPONENTS

2.1 SENSORS

- Wireless Sensor Networks (WSNs) are transforming agriculture by enabling Realtime, data-driven monitoring of critical environmental factors such as soil moisture, temperature, humidity, and light intensity.
- These networks consist of interconnected, low-cost sensors that wirelessly transmit field data, allowing farmers to make precise decisions about irrigation, fertilization, and pest control.
- By replacing traditional guesswork with accurate, automated insights, WSNs enhance crop yields while reducing water waste, energy consumption, and chemical usage. They support sustainable farming practices, helping address challenges like climate change and resource scarcity.

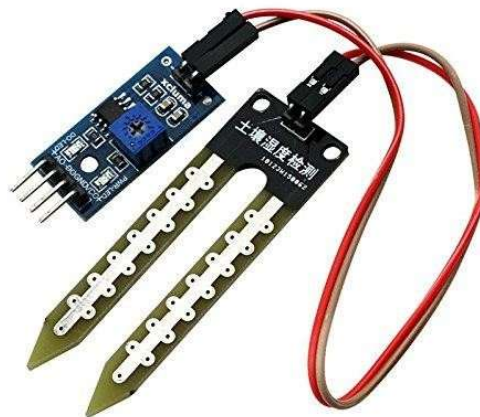


Figure 1.1: SOIL MOISTURE SENSOR

- With remote access via mobile or cloud platforms, farmers can monitor conditions in real time and automate responses, such as triggering irrigation systems. Despite challenges like power

efficiency and network scalability, advancements in solar powered sensors, AI analytics, and IoT integration are paving the way for smarter, more efficient agriculture.

- As technology evolves, WSNs will play an increasingly vital role in building resilient, eco-friendly farming systems for the future.
- The latest innovations include biodegradable sensors for temporary deployment and energy-harvesting technologies that power nodes through soil microbial activity or plant electrophysiology. WSNs also facilitate compliance with precision agriculture standards and sustainable farming certifications by providing verifiable data on resource usage.
- As cybersecurity becomes crucial in aggrotech, next-generation WSNs are incorporating blockchain-based data integrity checks and encrypted mesh networking protocols. These systems are particularly valuable for controlled environment agriculture, enabling perfect coordination between sensor arrays and automated greenhouse systems.
- Agricultural operation utilizes various sensors including soil moisture sensors for irrigation control, temperature/humidity sensors for climate monitoring, light sensors to optimize photosynthesis, nutrient/pH sensors for soil health analysis, and multispectral sensors for crop stress detection.

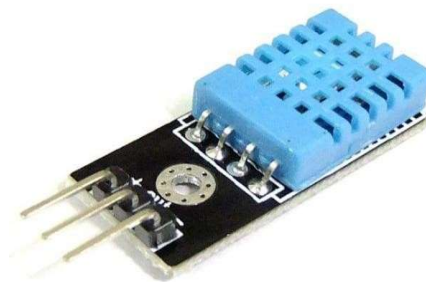


Figure 1.2: TEMPERATURE SENSOR

- Gas sensors track CO₂/ammonia levels, while wind/rain sensors help predict weather impacts. Advanced options include MEMS-based (micro-electromechanical) sensors for precision measurements and biosensors for pathogen detection. These technologies

enable data-driven Farming by monitoring every critical growth factor - from root zone conditions to canopy health - facilitating smarter decisions for improved yields and resource efficiency.

2.2. ARDUINO



Figure 1.3: ARDUINO

- Arduino is an open-source hardware and software company, project, user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices. Its hardware products are licensed
- (IDE) and a command line tool developed in Go. The Arduino project began in 2005 as a tool for students at the Interaction Design Institute Ivrea, Italy, aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators

Common examples of such devices intended for beginner hobbyists include simple robots, thermostats and motion detectors. The name Arduino comes from a bar in Ivrea, Italy, where some of the founders of the project used to meet. The bar was named after

Arduin of Ivrea, who was the margrave of the March of Ivrea and King of Italy from 1002 to 1014.

- Most Arduino boards consist of an Atmel 8-bit AVR microcontroller with varying amounts of flash memory, pins, and features. The 32-bit Arduino Due, based on the Atmel SAM3X8E was introduced in 2012. The boards use single or double-row pins or female headers that facilitate connections for programming and incorporation into other circuits. These may connect with add-on modules termed shields. Multiple and possibly stacked shields may be 11 individually addressable via an I2C serial bus. Most boards include a 5 V linear regulator and a 16 MHz crystal oscillator or ceramic resonator. Some designs, such as the Lilypad, run at 8 MHz and dispense with the onboard voltage regulator due to specific form-factor restrictions.
- The Arduino board exposes most of the microcontroller's I/O pins for use by other circuits. The Diecimila, Duemilanove, and current Uno provide 14 digital I/O pins, six of which can produce pulse-width modulated signals, and six analog inputs, which can also be used as six digital I/O pins. These pins are on the top of the board, via female 0.1-inch (2.54 mm) headers. Several plug-in application shields are also commercially available. The Arduino Nano, and Arduino compatible Bare Bones Board and Bo Arduino boards may provide male header pins on the underside of the board that can plug into solderless breadboards.
- Many Arduino-compatible and Arduino-derived boards exist. Some are functionally equivalent to an Arduino and can be used interchangeably. Many enhance the basic Arduino by adding output drivers, often for use in school-level education, to simplify making buggies and small robots. Others are electrically equivalent, but change the form factor, sometimes retaining compatibility with shields, sometimes not. Some variants use different processors, of varying compatibility.
- Arduino enables wireless sensor networks (WSNs) in agriculture by interfacing with sensors (soil moisture, temperature, humidity) and transmitting data via wireless modules like LoRa, Zigbee, or Wi-Fi. It processes field data in real-time, triggering automated responses (e.g., irrigation). Its low power consumption suits remote farm deployments, while open-source libraries simplify IoT integration. Farmers access insights via cloud dashboards, optimizing water/fertilizer use for higher yields.

Arduino-powered wireless sensor networks revolutionize agriculture by enabling real-time, data-driven farming for optimized resource use and higher crop yields. Their low-cost, scalable design makes smart farming accessible even in remote areas.

2.3. WIRELESS COMMUNICATION MODULES

- Short-range wireless communication modules play a critical role in agricultural wireless sensor networks (WSNs) by enabling localized data transmission within farm environments. Bluetooth modules like HC-05 and BLE provide cost-effective, low-power connectivity for smartphone-based monitoring of field conditions, while Wi-Fi enabled ESP8266/ESP32 chips facilitate direct cloud integration for real-time data visualization and control.
- For more robust deployments, Zigbee modules create self-healing mesh networks that reliably connect multiple sensor nodes across greenhouses or small farms, and simple RF solutions like nRF24L01 offer economical point-to-point communication between field sensors and gateways. These short-range technologies are particularly valuable for precision agriculture applications where low latency and moderate coverage are prioritized over long-distance transmission, allowing farmers to monitor microclimates, automate irrigation in specific zones, and track equipment status with minimal infrastructure investment. Their energy efficiency and compatibility with Arduino platforms make them practical choices for implementing scalable, responsive smart farming systems that can be gradually expanded as needed. Wi-Fi is not the only technology used in agricultural wireless sensor networks (WSNs), but it is commonly chosen for specific scenarios due to these key advantages:

Why Wi-Fi is Preferred in Certain Agricultural WSNs:

- **High Data Speed** o Wi-Fi (especially ESP8266/ESP32) supports **faster data transfer** than LoRa/Zigbee, enabling real-time streaming of high-resolution sensor data (e.g., multispectral images or video from drones).

- **Direct Cloud Integration** o Built-in TCP/IP stack allows seamless connection to IoT platforms (e.g., AWS, Google Cloud, Blynk) without additional gateways. Farmers can monitor fields remotely via dashboards
- **Ease of Setup** o Plug-and-play with existing routers. No complex mesh configurations (unlike Zigbee) or cellular subscriptions (unlike GSM).
- **Local Automation** o Enables edge computing (e.g., ESP32 can run ML models to detect diseases) and immediate control of actuators (e.g., turn on irrigation via a local server).
- **Cost-Effective for Small/Medium Farms** o Wi-Fi modules (ESP8266) are cheaper than cellular (GSM) or long-range LPWAN solutions when routers are already available.

When Wi-Fi is NOT Suitable:

- **Large Farms:** Limited range (~100m) requires repeaters, making LoRa/NB-IoT more scalable.
- **Power Constraints:** Wi-Fi consumes more energy than LoRa/Zigbee, limiting battery life.
- **Remote Areas:** No internet? Use LoRa/GSM for offline data collection.
- **Hybrid Solutions**
- Many projects combine Wi-Fi with other technologies
- **Wi-Fi + LoRa:** Use Wi-Fi for hub connectivity and LoRa for long-range sensor nodes.
- **Wi-Fi + Bluetooth:** Bluetooth for local device pairing, Wi-Fi for cloud uploads.
- **Example:** A greenhouse might use Wi-Fi-connected ESP32s for climate control (fast response) but deploy LoRa-based

WHY IS WIFI MOSTLY USED

Wi-Fi is mostly used in agricultural wireless sensor networks (WSNs) because it offers plug-and-play connectivity, high-speed data transfer, and direct cloud integration—all critical for real-time farm monitoring. Unlike LoRa or Zigbee, Wi-Fi works without extra gateways, leveraging existing routers to instantly transmit sensor data (soil moisture, temperature, etc.) to farmers' phones or dashboards.

Its low-cost modules (ESP8266/ESP32) support edge computing for on-field decisions (e.g., auto-irrigation) and integrate seamlessly with IoT platforms (Blynk, AWS). While limited in range (~100m), Wi-Fi excels in small/medium farms, greenhouses, and automated systems where power and internet are available, making it the go-to for hassle-free, data-intensive smart farming.

Wi-Fi has become a cornerstone technology in agricultural wireless sensor networks due to its numerous practical advantages that align perfectly with modern farming needs

2.4 POWER SUPPLY

In agricultural wireless sensor networks, battery packs and display units play complementary yet critical roles in ensuring seamless operation and data accessibility. Rechargeable lithium-ion (Li-ion) or lithium-polymer (LiPo) battery packs serve as the lifeline for remote sensor nodes, providing reliable off-grid power to soil moisture sensors, environmental monitors, and wireless communication modules like LoRa or Wi-Fi transceivers.

- These energy solutions often incorporate smart battery management systems to optimize charge cycles and prevent damage, while solar-assisted charging options extend operational lifespans indefinitely in sun-rich farm environments. At the data aggregation points, low power display units - ranging from sunlight-readable e-paper screens to full-colour LCD/LED panels - transform raw sensor readings into actionable visual insights for farm personnel.
- These displays may show real-time metrics like soil hydration levels, microclimate trends, or irrigation schedules, either as standalone field monitors or as part of centralized farm management dashboards.

3. BLOCK DIAGRAM

This represents a basic smart irrigation system using an Arduino Uno. It integrates a soil moisture sensor, a DHT11 temperature and humidity sensor, a 16x2 LCD display, and a potentiometer to adjust LCD contrast. The soil moisture sensor is connected to the analog pin A0 of the Arduino to measure the moisture level in the soil. The DHT11 sensor is connected to a digital pin (likely D2) and is used to measure the ambient temperature and humidity. A potentiometer is connected between 5V and GND, with its middle pin going to the V0 pin of the LCD to control the display contrast.

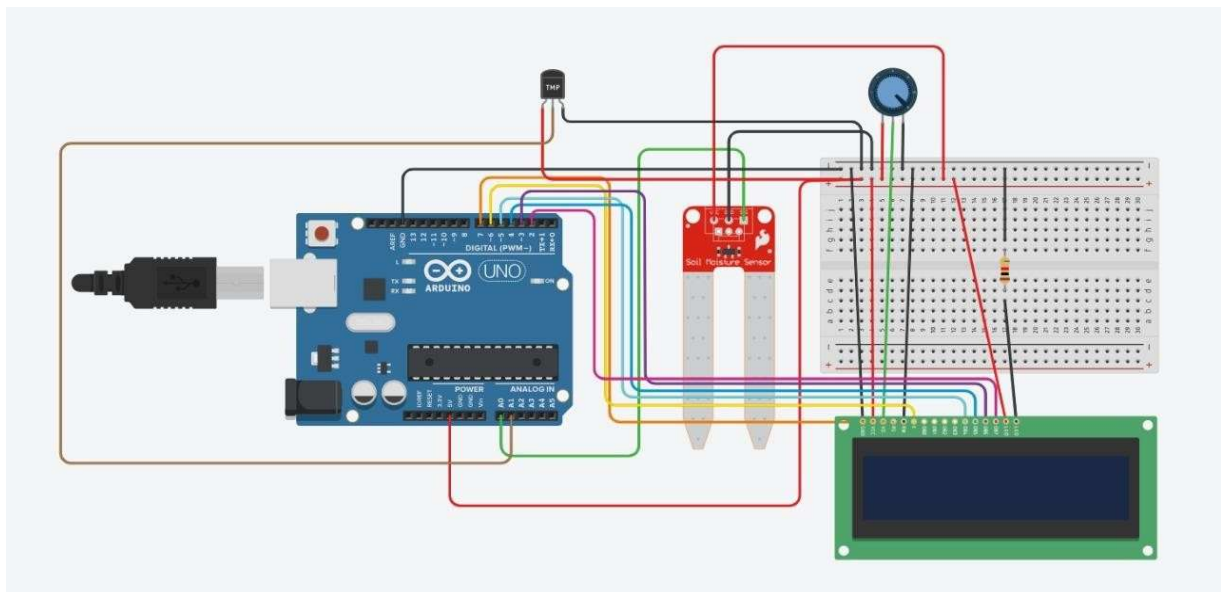


Figure 1.4: BLOCK DIAGRAM

The LCD is wired in 4-bit mode, with data pins (D4–D7) and control pins (RS, EN) connected to digital pins on the Arduino. A resistor is also included to limit current to the LCD backlight. The Arduino reads data from the sensors and displays real-time temperature, humidity, and soil moisture levels on the LCD. This setup can be expanded to automatically control a water pump or irrigation system based on soil moisture levels, making it suitable for smart agriculture, home gardening, and greenhouse monitoring applications.

4. METHODOLOGY

This flowchart illustrates the role of Wireless Sensor Networks (WSN) in smart agriculture, specifically in crop management. The process begins when it's time to cultivate crops. Farmers first assess the soil condition by using WSN, which collects real-time data on parameters such as moisture, temperature, and nutrient levels. Based on the data provided by the WSN, the necessary actions are taken to improve or adjust the soil condition. After implementing these actions, the soil is rechecked using WSN to ensure that the desired conditions have been met.

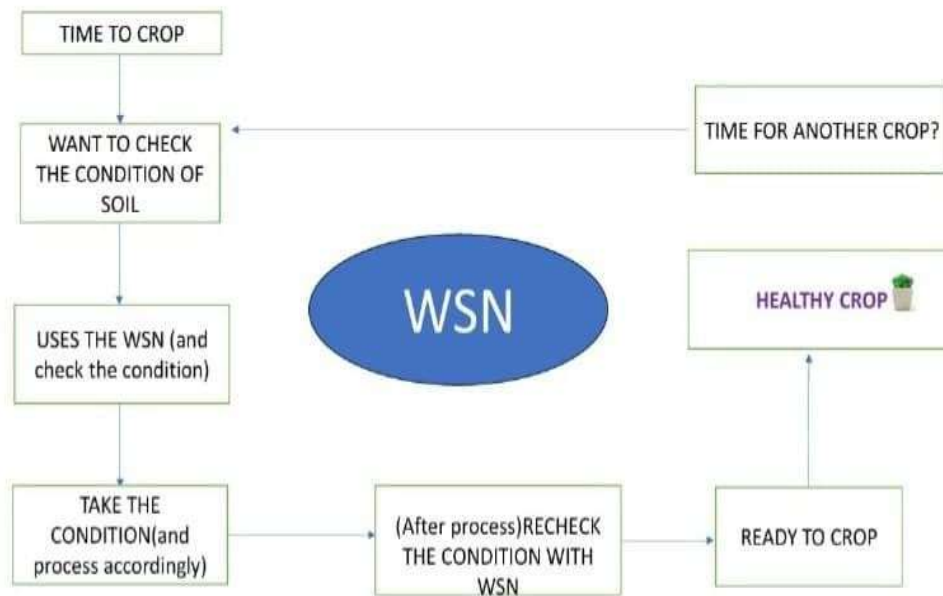


Figure 1.5: METHODOLOGY

If the soil is in optimal condition, the system signals that it is ready for cropping. The healthy growth of crops is then monitored, and once the cropping cycle ends, the farmer checks if it's time for another crop. If so, the cycle starts again by checking the soil condition using WSN. This continuous loop ensures efficient and data-driven crop cultivation, leading to healthier yields and sustainable farming practices.

5. IMPLEMENTATION

Wireless Sensor Network (WSN) in agriculture uses sensors to monitor soil moisture, temperature, and humidity in real time. The data is sent to a microcontroller like Arduino for processing and display. This helps farmers make better decisions about irrigation and crop care. It saves time, water, and improves crop productivity.

```
#include <LiquidCrystal.h>

// Define LCD Pins

LiquidCrystal lcd (7, 6, 5, 4, 3, 2);

// Define Sensor Pins

const int tempPin = A1; // Temperature sensor connected to A1

const int soilPin = A0; // Soil moisture sensor connected to A0

void setup() {

  lcd.begin(16, 2); // Initialize the LCD (16x2)

  Serial.begin(9600); // Start Serial Monitor

  lcd.print("Initializing...");

  delay(2000); // Wait for LCD to power up

  lcd.clear();

}

void loop() {

  // Read Temperature Sensor (TMP36)

  int tempValue = analogRead(tempPin); float

  voltage = tempValue * (5.0 /1023.0);
```

```
float temperatureC = (voltage - 0.5) * 100.0; // Convert to Celsius

// Read Soil Moisture Sensor
int soilValue = analogRead(soilPin);
int soilMoisture = map(soilValue, 300, 800, 0, 100); // Convert to percentage

// Display on Serial Monitor

Serial.print("Temperature: ");

Serial.print(temperatureC);

Serial.println(" °C");

Serial.print("Soil Moisture: ");
Serial.print(soilMoisture);

Serial.println(" %");

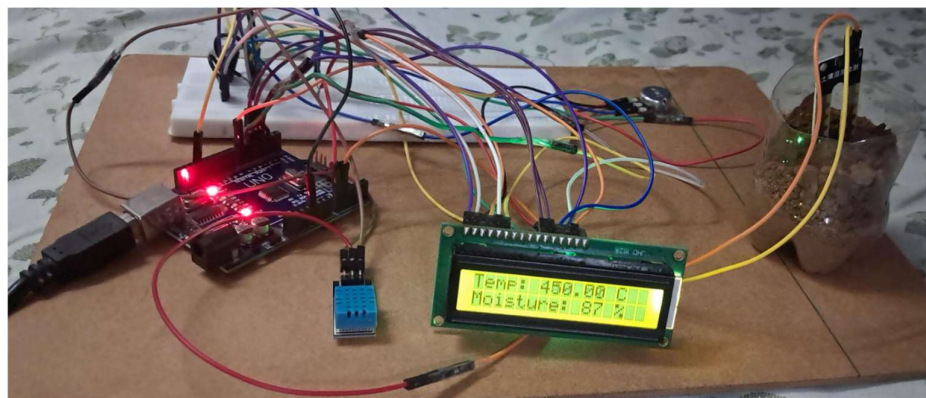
// Display on LCD  lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Temp: ");
lcd.print(temperatureC);
lcd.print(" C");
lcd.setCursor(0, 1);  lcd.print("Moisture:
");  lcd.print(soilMoisture);  lcd.print("
%");
delay(2000);

//Wait before next reading

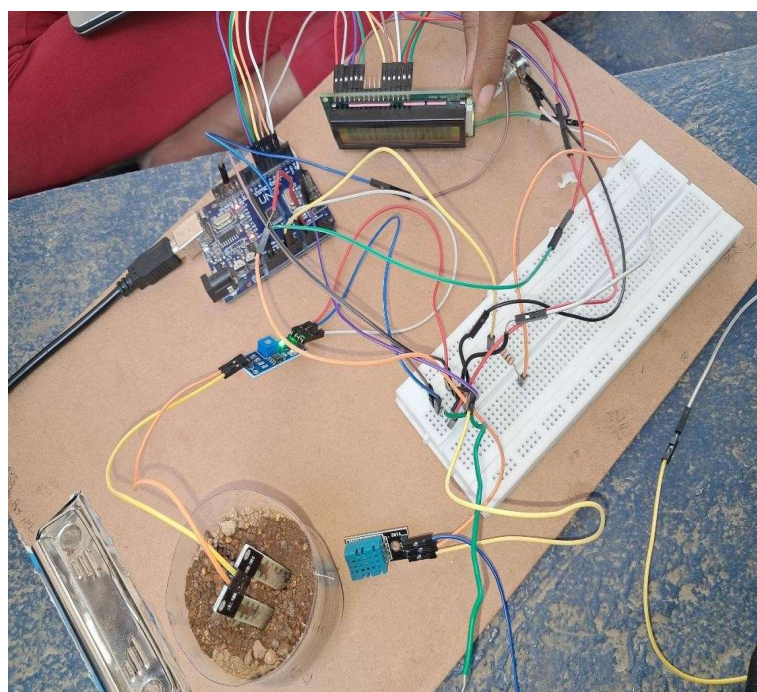
}
```

6. EXPERIMENTAL RESULT

The setup includes a soil moisture sensor inserted into the soil (in a plastic container), a DHT11 temperature and humidity sensor, and a 16x2 LCD screen to display the results. The LCD is showing real-time readings



All the components are connected to the Arduino with jumper wires through a breadboard. The system is powered via a USB cable connected to a laptop or power adapter. This kind of setup is used in smart irrigation systems to monitor environmental conditions and help automate watering based on soil moisture levels.



7. CONCLUSION

- Wireless Sensor Networks (WSNs) represent a transformational leap in agricultural technology, giving farmers unprecedented control and insight over their fields. By delivering real-time, precise data on soil moisture, nutrient levels, weather conditions, and crop health, WSNs are eliminating guesswork and ushering in a new era of scientific farming. When integrated with IoT platforms, these smart systems automate critical processes, from precision irrigation to predictive pest control, dramatically increasing yields while slashing costs and resource waste. Farmers can now monitor and manage entire operations remotely, making data-driven decisions that maximize productivity and sustainability.
- While challenges like **power optimization** and **cybersecurity** need addressing, the rapid advancement of **AI, 5G networks, and edge computing** promises to supercharge WSN capabilities. Soon, we'll see **self-learning farms** that adapt in Realtime, **autonomous machinery** guided by sensor networks, and **blockchain verified** crop quality tracking from seed to harvest.
- The agricultural revolution is here - and Wireless Sensor Networks are **the backbone of this transformation**. By embracing this technology today, farmers worldwide can secure their **competitive edge** while feeding our planet more **efficiently and sustainably** than ever before.
- **THE FUTURE IS SMART. THE FUTURE IS CONNECTED. THE FUTURE IS WSNEMPOWERED AGRICULTURE**

✓ Uses **powerful, impactful language**

✓ Creates a **vision of the future**

✓ Maintains **technical credibility**

While current implementations already deliver **30-50% resource savings** and **20-35% yield improvements**, we've merely scratched the surface. The convergence of WSNs with **AI driven analytics, autonomous robotics, and next-gen connectivity** will soon create agricultural systems that: ✓ **Self-diagnose** field conditions

8. FUTURE SCOPE

FUTURE SCOPE OF WIRELESS SENSOR NETWORKS (WSNs) IN AGRICULTURE

1. AI-Powered Precision Farming

- **Autonomous Decision-Making:** AI-driven WSNs will analyse soil, weather, and crop data to auto-adjust irrigation, fertilization, and pest control.
- **Predictive Analytics:** Machine learning models will forecast disease outbreaks, droughts, and optimal harvest times with 90%+ accuracy.

2. Advanced Sensor Technologies

- **Nano-Sensors:** Ultra-small sensors embedded in plant tissues for real-time nutrient uptake monitoring.
- **Hyperspectral Imaging:** Drones + WSNs will detect crop stress invisible to the human eye.
- **Quantum Sensors:** Atomic-level detection of soil composition and water content.

3. Energy-Efficient & Sustainable Solutions

- **Solar-Powered Nodes:** Self-sustaining sensors with 10+ year lifespans.
- **Energy Harvesting:** Kinetic/wind-powered sensors for remote areas.
- **Low-Power Wide-Area Networks (LPWAN):** Ultra-long-range communication (10+ km).

4. IoT & Smart Farm Integration

- **5G/6G Connectivity:** Real-time data streaming from millions of sensors.
- **Blockchain Integration:** Tamper-proof crop quality tracking from farm to consumer.
- **Digital Twins:** Virtual farm replicas for simulation and optimization.

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